

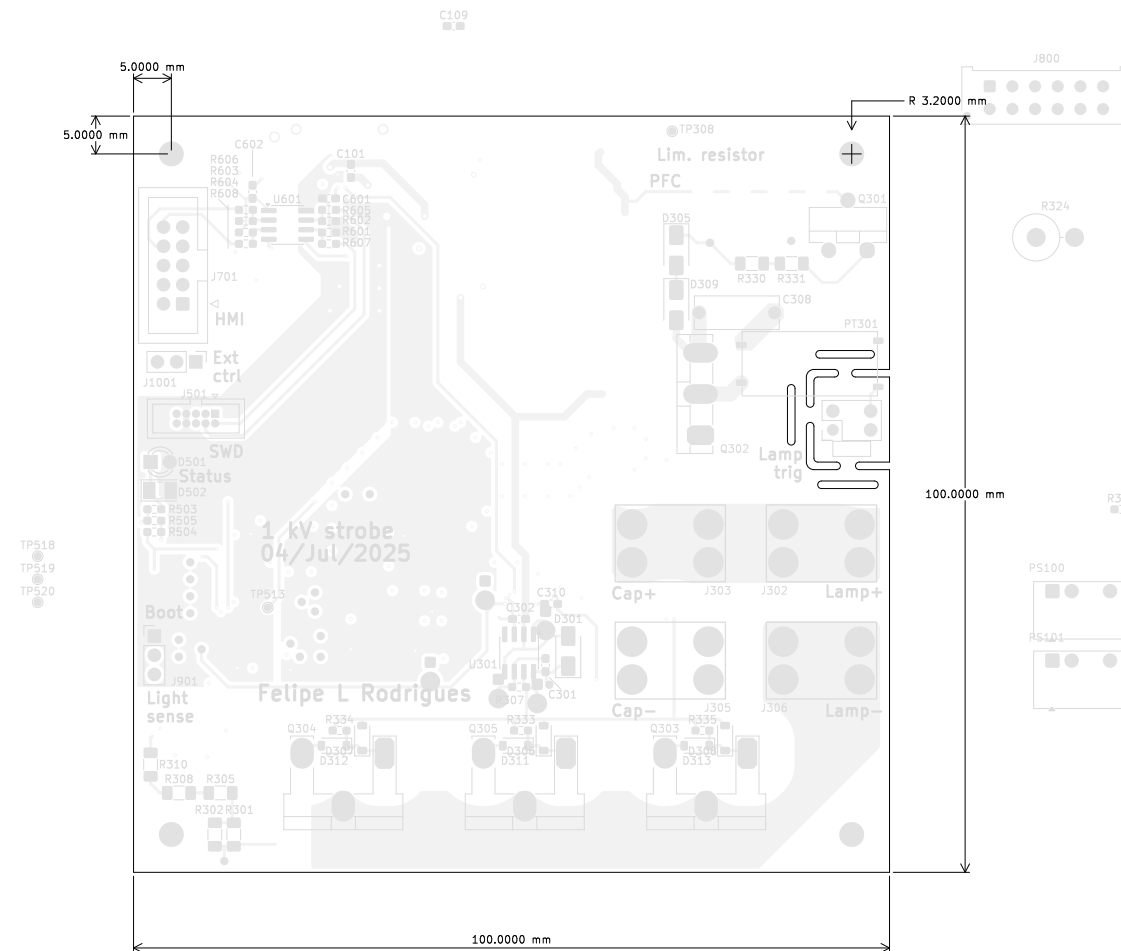
Main board Fabrication Document

Top Fabrication (Scale 1:1)

Layer Stack Legend

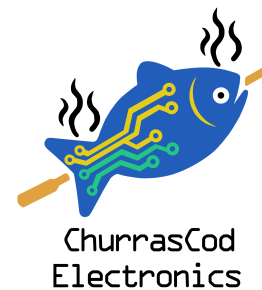
	Material	Layer	Thickness	Dielectric	Type	Gerber
		F.Paste			Paste Mask	
		F.Silkscreen		Direct Printing	Legend	GBR
		F.Mask	0.02mm	Solder Resist	Solder Mask	GBR
	Copper	F.Cu (Sig, PWR)	0.07mm (2.00oz)		Signal	GBR
	Prepreg		0.1mm	FR4_7628	Dielectric	
	Copper	In1.Cu (GND)	0.035mm (1.00oz)		Plane	GBR
	Core		1.828mm	FR4	Dielectric	
	Copper	In2.Cu (Sig, PWR)	0.035mm (1.00oz)		Plane	GBR
	Prepreg		0.1mm	FR4_7628	Dielectric	
	Copper	B.Cu (Sig, PWR)	0.07mm (2.00oz)		Signal	GBR
		B.Mask	0.02mm	Solder Resist	Solder Mask	GBR
		B.Silkscreen		Direct Printing	Legend	GBR
		B.Paste			Paste Mask	

Total thickness: 2.278mm
Note: external layer thicknesses are specified after plating



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.
- DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.
- SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 4) SURFACE FINISH: HAL SNPB
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR GREEN.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:
- A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
- B. Tg 170 C OR EQUIVALENT.
- C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY CHURRASCOD ELECTRONICS.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:
- | | |
|------------------|----------------------|
| BOARD SIZE | 100,000 x 100,000 mm |
| BOARD THICKNESS | 2,278 mm |
| TRACE WIDTH | 0,300 mm |
| TRACE TO TRACE | 0,200 mm |
| MIN. HOLE (PTH) | 0,300 mm |
| MIN. HOLE (NPTH) | 1,000 mm |
| ANNULAR RING | 0,100 mm |
| COPPER TO HOLE | 0,300 mm |
| COPPER TO EDGE | 0,500 mm |
| HOLE TO HOLE | 0,250 mm |



Comments:

Company:

ChurrasCod Electronics

Board Name:

Main board

File Name:

Strobe_main.kicad_pcb

Variant:

PRELIMINARY

Git Hash:

c741cc4

Project Name:

1 kV strobe

Date:

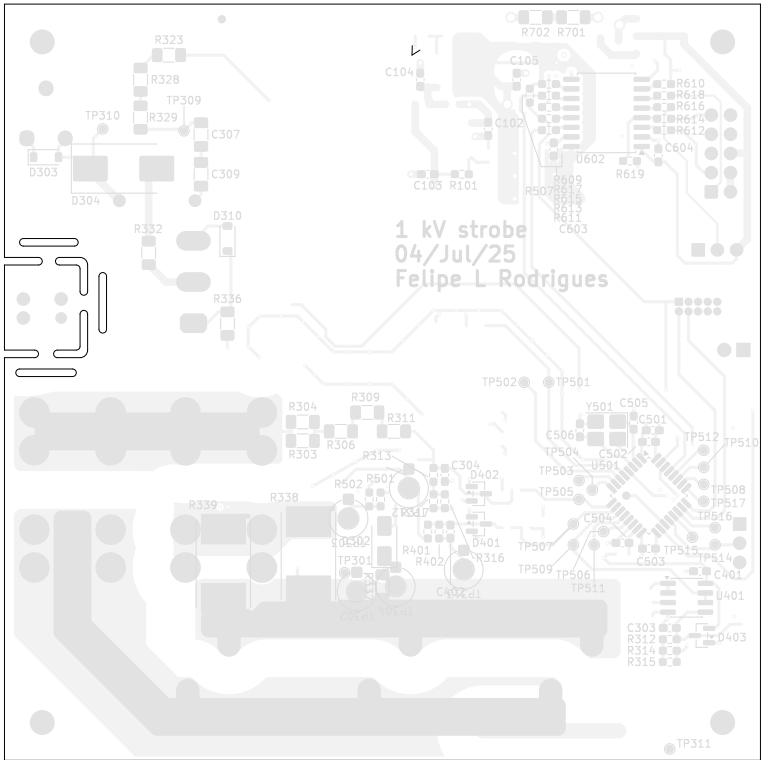
2024-04-13	1.0.0+ (Unreleased)
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Size:

A3 1 of 10

Main board Fabrication Document

Bottom Fabrication (Scale 1:1)



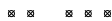
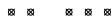
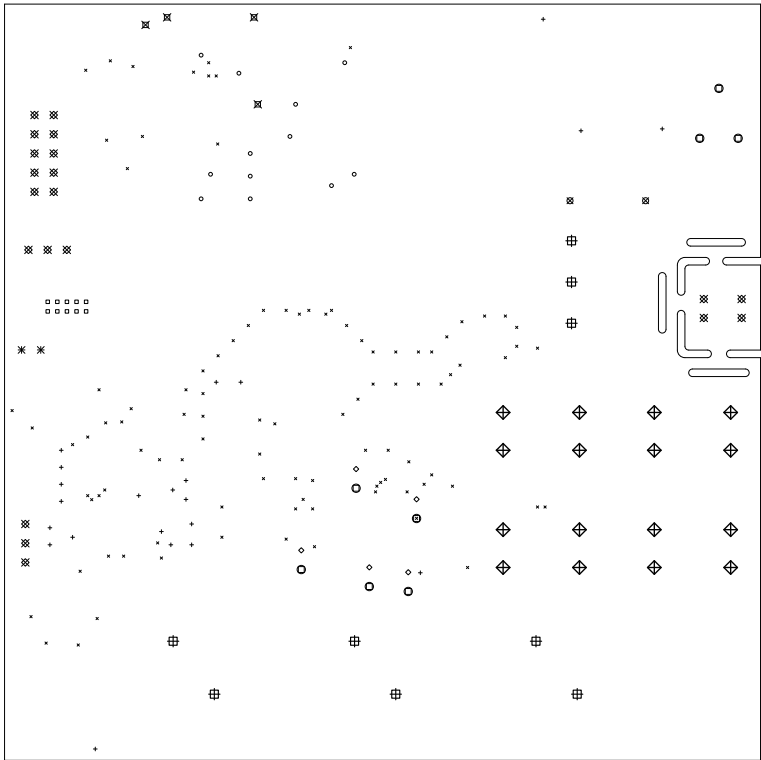
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			Board Name: Main board		Project Name: 1 kV strobe	
	Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)	
Sheet Path:			Reviewer:	Size: A3	Sheet: 2 of 10	

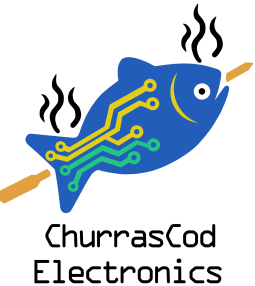
Main board Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	101	0,30mm (11,81mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Via
○	12	0,50mm (19,69mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Via
+	25	0,50mm (19,69mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
□	10	0,65mm (25,59mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
◇	5	0,70mm (27,56mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊗	12	0,80mm (31,50mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
※	2	0,90mm (35,43mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊠	4	1,00mm (39,37mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Via
⊗	20	1,00mm (39,37mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊕	12	1,02mm (40,16mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
○	8	1,10mm (43,31mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
○	2	1,20mm (47,24mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊕	9	1,50mm (59,06mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊕	16	1,80mm (70,87mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
Total 238						



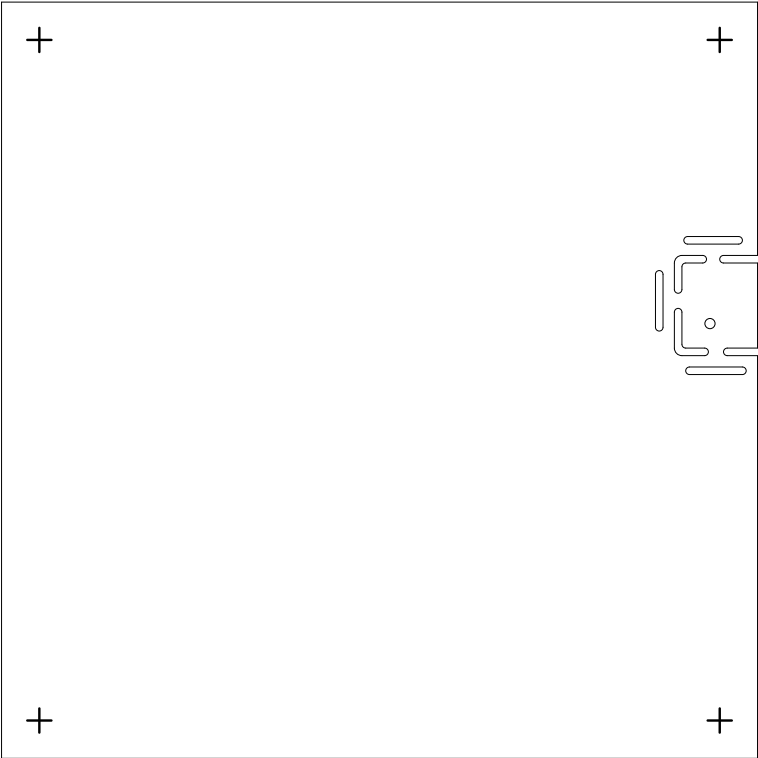
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			Board Name: Main board		Project Name: 1 kV strobe	
	Sheet Title: Drill Drawing (L1 - L4)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)	
	Sheet Path:		Reviewer:	Size: A3	Sheet: 3 of 10	

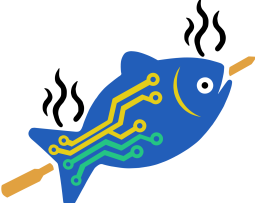
Main board Fabrication Document

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	2	1,02mm (40,16mils)	NPTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Mechanical
○	1	1,35mm (53,15mils)	NPTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Mechanical
+	4	3,20mm (125,98mils)	NPTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Mechanical
Total 7						

Drill Drawing L1 - L4 (Scale 1:1)





ChurrasCod Electronics

Comments:

Sheet Title:
Drill Drawing (L1 - L4)

Sheet Path:

Company:
ChurrasCod Electronics

Board Name:
Main board

File Name:
Strobe_main.kicad_pcb

Designer:
FR

Reviewer:

Variant:
PRELIMINARY

Project Name:
1 kV strobe

Date:
2024-04-13

Size:
A3

Git Hash:
c741cc4

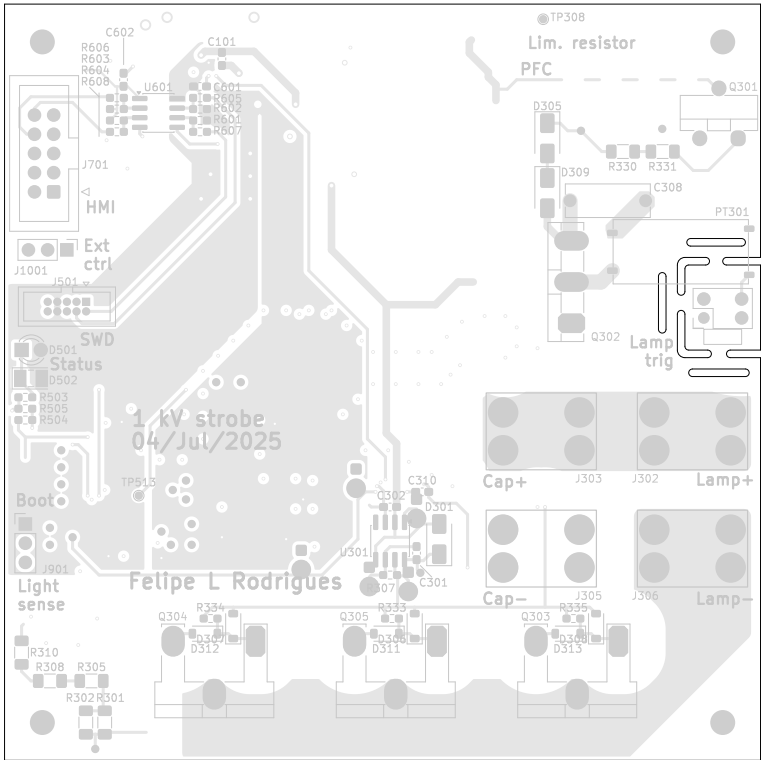
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Sheet:
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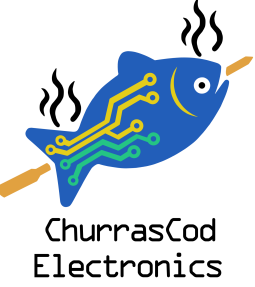
Main board Fabrication Document

Top Test Points (Scale 1:1)

TP518
TP519
TP520



Ref.	Net	X [mm]	Y [mm]
TP308	S1k	76.25	103.00
TP513	Vbus_ctrl	22.75	40.00
TP518	BOOT0	-7.68	46.80
TP519	SH0	-7.68	43.75
TP520	SH0	-7.68	40.70

	Comments:		Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: c741cc4
	Sheet Title: Top Test Points (Scale 1:1)		Board Name: Main board		Project Name: 1 kV strobe	
	Sheet Path:		File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
			Reviewer:		Size: A3	Sheet: 5 of 10

Main board Fabrication Document

Bottom Test Points (Scale 1:1)

A

B

C

D

E

F

A

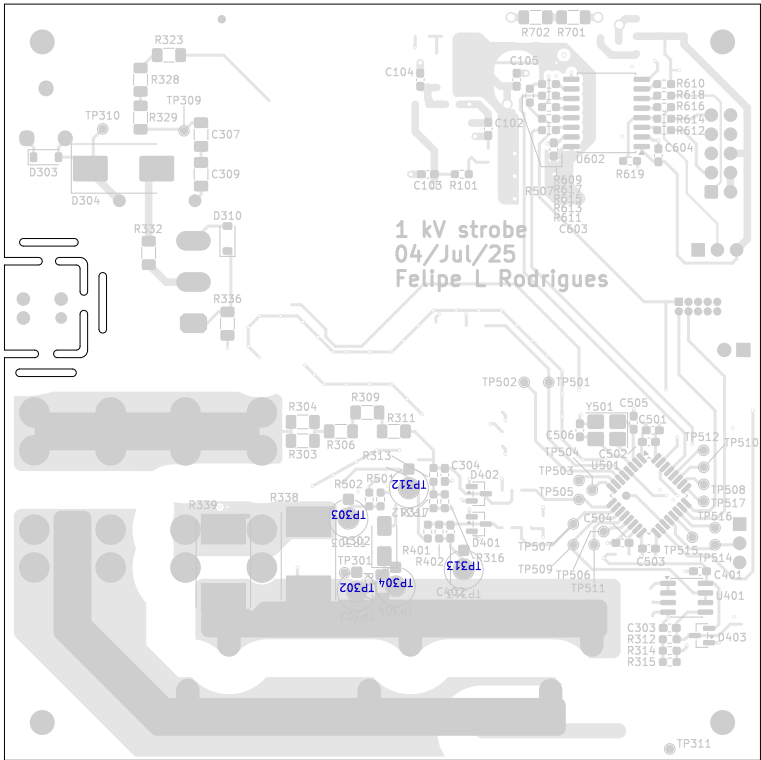
B

C

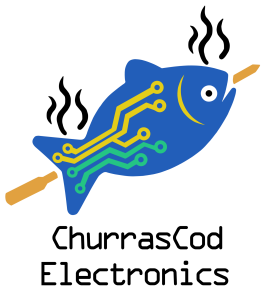
D

E

F



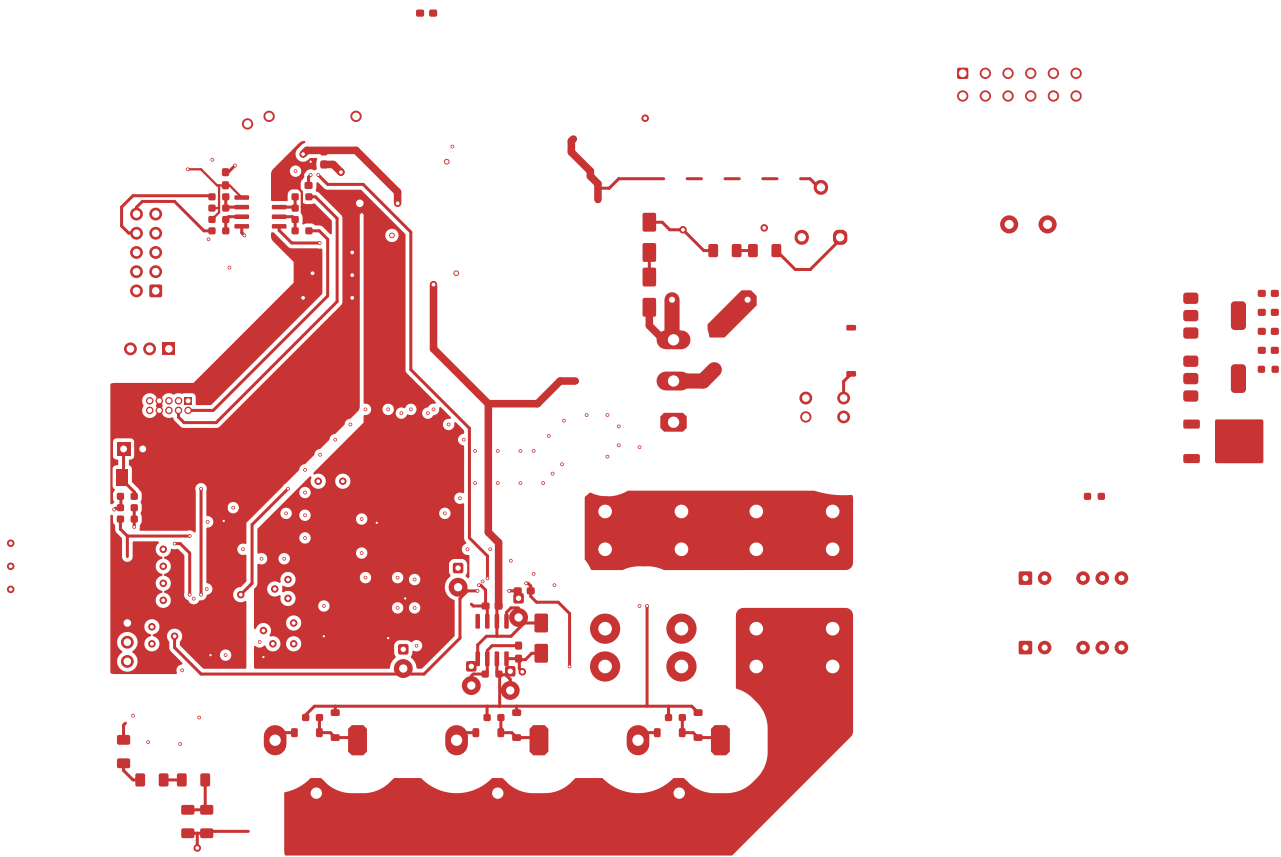
Ref.	Net	X [mm]	Y [mm]
TP301	VBS	60.00	29.80
TP302	SH0	58.40	28.60
TP303	SH0	59.50	38.25
TP304	Mon_S1k	53.25	29.25
TP309	TP_REF	81.25	88.25
TP310	TP_PRI	92.00	88.50
TP311	LMP	17.00	6.50
TP312	Mon_IL	51.50	42.25
TP313	DRV_OUT	44.25	31.50
TP501	Net-(U501-PA0)	33.00	55.00
TP502	Net-(U501-PA1)	36.25	55.00
TP503	SUP_TX	29.00	42.00
TP504	SUP_RX	27.25	40.75
TP505	Mon_S1k_buf	29.00	39.50
TP506	Net-(U501-PB0)	25.75	35.25
TP507	Mon_IL_prot	29.75	36.25
TP508	Net-(U501-PB3)	12.50	41.50
TP509	Mon_LMP_buf	29.75	33.50
TP510	Net-(U501-PB4)	12.50	43.75
TP511	Net-(U501-PA7)	27.00	33.50
TP512	Net-(U501-PB5)	12.50	46.00
TP514	Light_sns	11.00	33.50
TP515	INT_EN	14.00	34.50
TP516	Net-(U501-PA12)	11.00	35.75
TP517	Net-(U501-PA15)	12.50	39.25

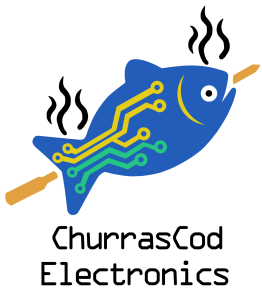


Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: c741cc4
	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: Bottom Test Points (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 6 of 10

Main board Fabrication Document

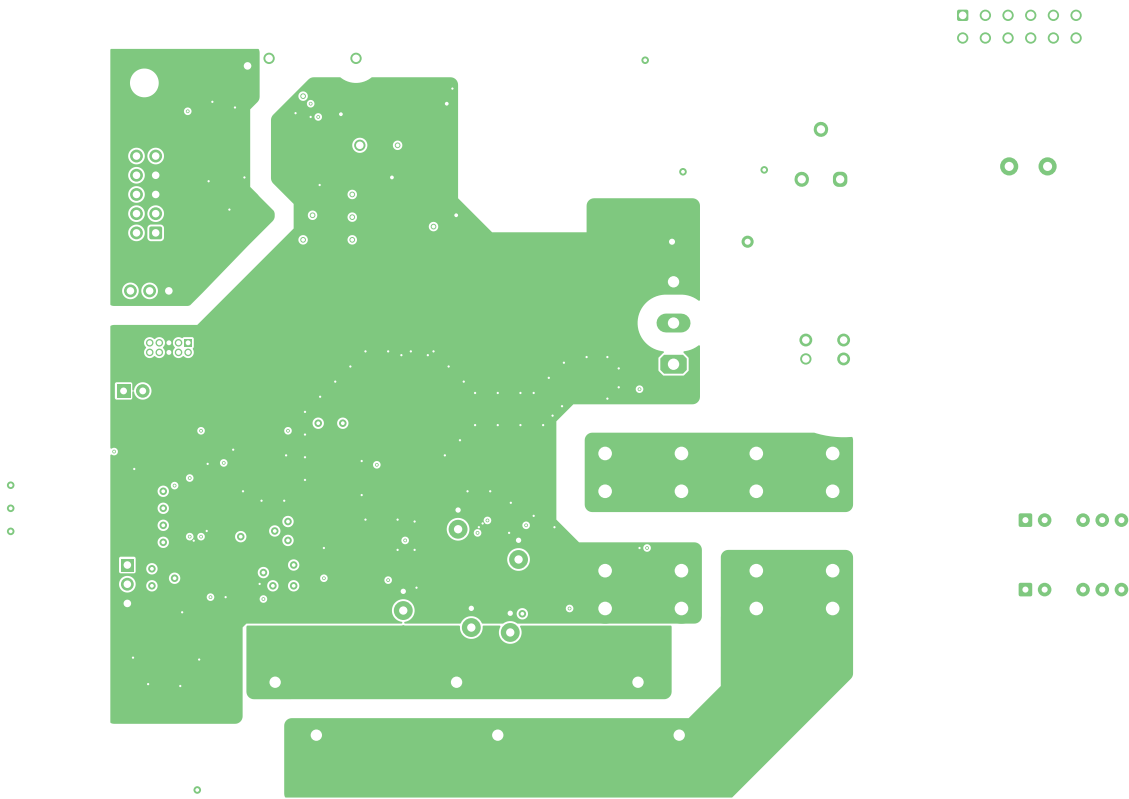
F.Cu (Sig, PWR) (Scale 1:1)

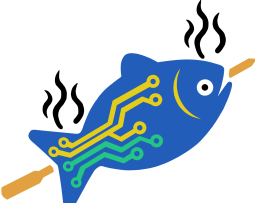


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	Sheet Title: F.Cu (Sig, PWR) (Scale 1:1)		Board Name: Main board		Project Name: 1 kV strobe	
	Sheet Path:		File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
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Main board Fabrication Document

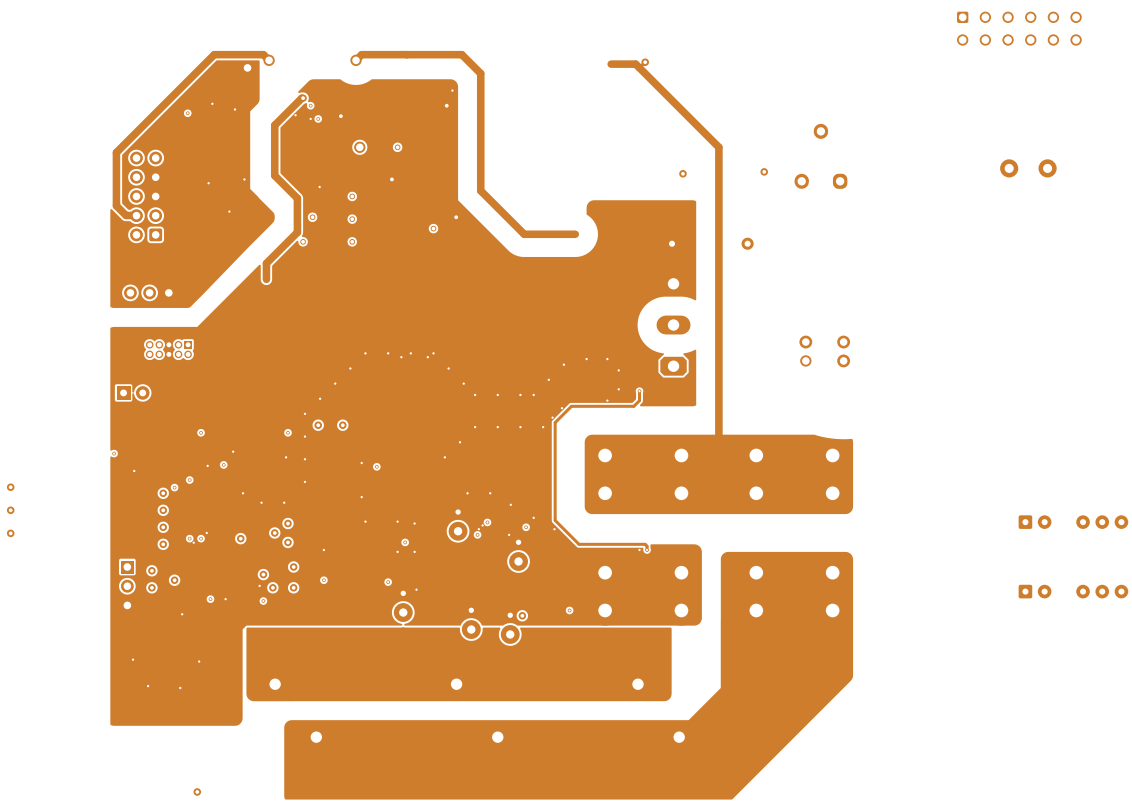
In1.Cu (GND) (Scale 1:1)

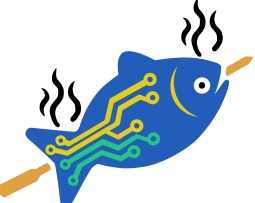


 ChurrasCod Electronics	Comments:		Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: c741cc4
			Board Name: Main board		Project Name: 1 kV strobe	
	Sheet Title: In1.Cu (GND) (Scale 1:1)	File Name: Strobe_main.kicad_pcb		Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
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Main board Fabrication Document

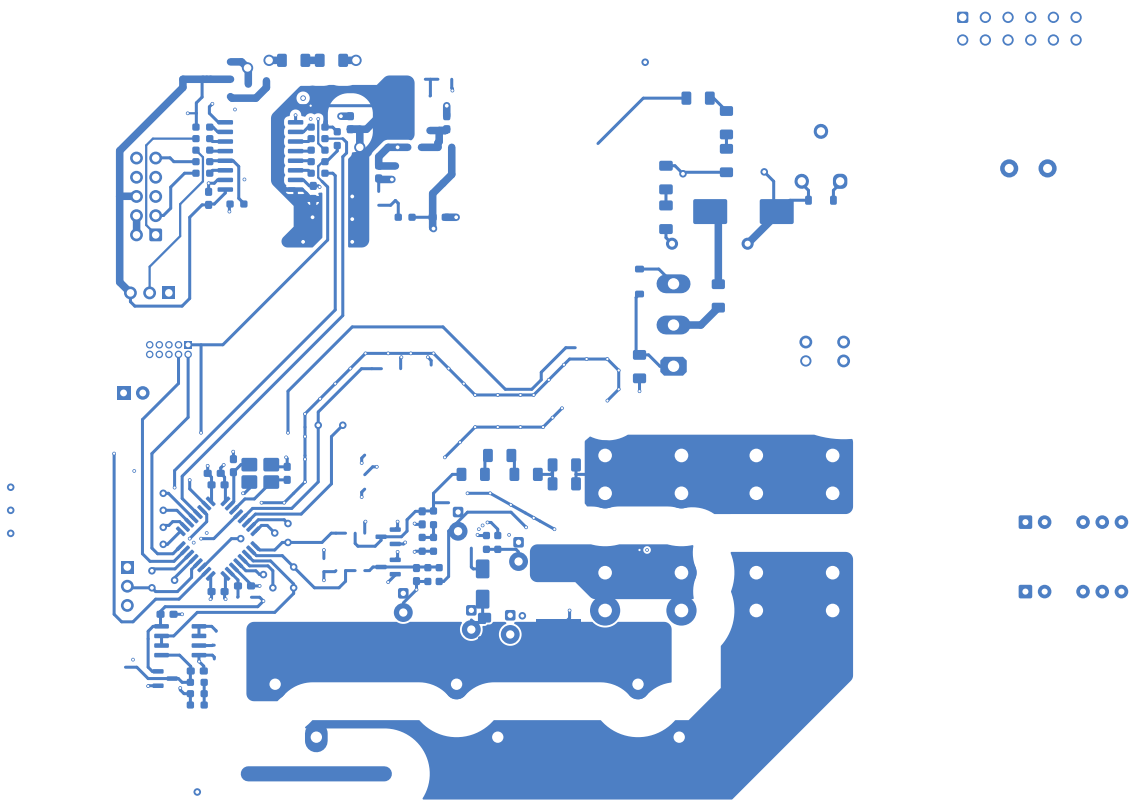
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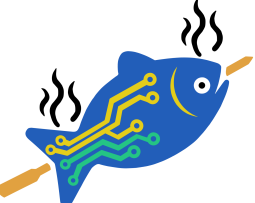


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	Sheet Title: In2.Cu (Sig, PWR) (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
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Main board Fabrication Document

B.Cu (Sig, PWR) (Scale 1:1)



 ChurrasCod Electronics	Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: c741cc4
	Sheet Title: B.Cu (Sig, PWR) (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
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